

Mean Absolute deviation: of a sample $\{x_1, x_2, \dots, x_n\}$ of size n is defined as $\frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$

If $x_1, x_2, \dots, x_p$ occurs with corresponding frequencies $f_1, f_2, \dots, f_p$, then mean absolute deviation $= \frac{1}{n} \sum_{i=1}^p f_i |x_i - \bar{x}|$, where $\sum_{i=1}^p f_i = n$.

Moments and measures of skewness and kurtosis:

For a sample $\{x_1, x_2, \dots, x_n\}$ of size n , the r th moment about origin, also called the r th raw moment denoted by m_r' , is defined as $m_r' = \frac{1}{n} \sum_{i=1}^n x_i^r$, where r is a non-negative integer

The r th moment about the mean $\bar{x}$, also called the r th central moment denoted by m_r is defined as

$$m_r = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r.$$

Clearly, $m_1' = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$.

Also, $m_1 = 0$ for all frequency distributions.

The r th moment about an arbitrary origin a is also denoted by m_r' , is defined as $m_r' = \frac{1}{n} \sum_{i=1}^n (x_i - a)^r$.

If the sample values occur with frequencies $f_1, f_2, \dots, f_n$, where $\sum_{i=1}^n f_i = N$, then the r th moment about origin is

$$m_r' = \frac{1}{N} \sum_{i=1}^n f_i x_i^r$$

Also, the r th moment about the mean is given by

$$m_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r$$

The n th moment about an arbitrary origin a is given by

$$m'_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - a)^r.$$

- o The relation between the n th central moment (i.e., the n th moment about mean) m_r and moments about origin of order less than or equal to r is given by the following:

$$\begin{aligned} m_2 &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 \\ &= \frac{1}{N} \sum_{i=1}^n f_i x_i^2 - 2\bar{x} \frac{1}{N} \sum_{i=1}^n f_i x_i + \bar{x}^2 \\ &= m'_2 - \bar{x}^2 = m_2 - m_1'^2 \end{aligned}$$

$$\begin{aligned} m_3 &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^3 \\ &= \frac{1}{N} \sum_{i=1}^n f_i x_i^3 - 3\bar{x} \frac{1}{N} \sum_{i=1}^n f_i x_i^2 + 3\bar{x}^2 \frac{1}{N} \sum_{i=1}^n f_i x_i - \bar{x}^3 \\ &= m'_3 - 3\bar{x} m'_2 + 3\bar{x}^3 - \bar{x}^3 \\ &= m'_3 - 3m'_2 m'_1 + 2m_1'^3 \end{aligned}$$

Similarly,

$$m_4 = m'_4 - 4m'_3 m'_1 + 6m'_2 m_1'^2 - 3(m_1')^4$$

Ex: For a distribution of 280 observations, the mean is 54 and standard deviation is 3. On checking, it was observed that observations which should have correctly read as 62 and 82 had been wrongly recorded as 64 and 80 respectively. Calculate the correct values of mean and standard deviation.

Sol: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, $n = 280$

Using the incorrect mean,

$$\sum_{i=1}^n x_i = 54 \times 280$$

which includes the incorrect observations 64 and 80.

$$\begin{aligned} \text{Hence the correct sum} = \sum x_i &= 54 \times 280 - (64 + 80) + 62 + 82 \\ &= 280 \times 54 \end{aligned}$$

$$\text{Therefore the correct mean} = \frac{280 \times 54}{280} = 54$$

$$\text{Again, } \frac{1}{n} \sum x_i^2 - \bar{x}^2 = s^2 = 9$$

$$\text{or, } \sum x_i^2 = (9 + 54^2) 280 = 819000$$

$$\begin{aligned} \text{So the correct value } \sum x_i^2 &= 819000 - (64^2 + 80^2) + 62^2 + 82^2 \\ &= 819072 \end{aligned}$$

$$\text{Therefore the correct variance} = \frac{819072}{280} - 54^2 = 9.26 \text{ (approx.)}$$

$$\text{Hence the correct standard deviation} = \sqrt{9.26} = 3.04$$

$$\begin{aligned} s^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 \end{aligned}$$

Ex: The arithmetic mean of a distribution is 5, while the 2nd and 3rd central moments are 20 and 140 respectively. Find the first three moments of the distribution about 10.

Sol: Given that $\frac{1}{n} \sum_{i=1}^n x_i = \bar{x} = 5$,

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = 20$$

and

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3 = 140$$

$$\text{First moment about 10} = \frac{1}{n} \sum_{i=1}^n (x_i - 10) = \frac{1}{n} \sum_{i=1}^n (x_i - 5 - 5)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - 5) - 5$$

$$= m_1 - 5 = -5 \quad (\because m_1 = 0)$$

$$\text{Second moment about 10} = \frac{1}{n} \sum_{i=1}^n (x_i - 10)^2 = \frac{1}{n} \sum_{i=1}^n (x_i - 5 - 5)^2$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - 5)^2 - 10 \cdot \frac{1}{n} \sum_{i=1}^n (x_i - 5) + 25$$

$$= 20 + 25 = 45$$

$$\text{Third moment about 10} = \frac{1}{n} \sum_{i=1}^n (x_i - 10)^3 = \frac{1}{n} \sum_{i=1}^n (x_i - 5 - 5)^3$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - 5)^3 - 15 \cdot \frac{1}{n} \sum_{i=1}^n (x_i - 5)^2 + 75 \cdot \frac{1}{n} \sum_{i=1}^n (x_i - 5) - 125$$

$$= 140 - 15 \times 20 - 125 = -285$$

Ex: The scores of two batsman A and B in ten innings during a certain season are as follows:

A	32	28	47	63	71	39	10	60	96	14
B	19	31	48	53	67	90	10	62	40	80

Find which of the two batsman, A and B, is more consistent in scoring.

Sol:

Score of A (x_i)	$u_i = x_i - x_0$ $= x_i - 46$	u_i^2	Score of B (y_i)	$v_i = y_i - y_0$ $= y_i - 50$	v_i^2
32	-14	196	19	-31	961
28	-18	324	31	-19	361
47	1	1	48	-2	4
63	17	289	53	3	9
71	25	625	67	17	289
39	-7	49	90	40	1600
10	-36	1296	10	-40	1600
60	14	196	62	12	144
96	50	2500	40	-10	100
14	-32	1024	80	30	900
Total	0	6500	0	0	5988

$$\bar{u} = \frac{\sum u_i}{n} = 0, \quad \bar{x} = x_0 + \bar{u} = 46$$

$$S.D. = S_x = S_u = \sqrt{\frac{\sum u_i^2}{n} - \bar{u}^2} = \sqrt{\frac{6500}{10} - 0} = 25.5$$

$$\text{Coefficient of variation} = \frac{S_x}{\bar{x}} = \frac{25.5}{46} = 0.5543$$

$$\bar{v} = \frac{\sum v_i}{n} = 0, \quad \bar{y} = y_0 + \bar{v} = 50$$

$$S.D. = S_y = S_v = \sqrt{\frac{5988}{10} - 0} = \sqrt{598.8} = 24.47$$

$$\text{Coefficient of variation} = \frac{S_y}{\bar{y}} = \frac{24.47}{50} = 0.4898$$

Since coefficient of variation of the scores of B is less than that of A, batsman-B is more consistent in scoring.

Ex: Let the mean and standard deviation of a sample of size n_1 with values x_i ($i=1, 2, \dots, n_1$) be denoted by $\bar{x}_1$ and s_1 . Similarly, for another sample of size n_2 with values x_2 ($i=1, 2, \dots, n_2$) by $\bar{x}_2$ and s_2 . Then the variance s^2 of the composite sample is given by

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} + \frac{n_1 n_2 (\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2}$$

Sol: Let $\bar{x}$ be the mean of the composite set. Now $\bar{x}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i$

$$\text{or, } \sum_{i=1}^{n_1} x_i = n_1 \bar{x}_1$$

$$\text{Also } \bar{x}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_{2i} \quad \text{or, } \sum_{i=1}^{n_2} x_{2i} = n_2 \bar{x}_2$$

Ex: For a distribution the mean is 10, variance is 16, γ_2 is +1 and β_2 is 4. Obtain the first four moments about the origin. Comment upon the nature of distribution.

Sol: We are given mean = 10, $\mu_2 = 16$, $\gamma_2 = +1$, $\beta_2 = 4$

First four moments about the origin (m_1', m_2', m_3', m_4')

$m_1' =$ first moment about the origin = mean = 10.

$$m_2 = m_2' - m_1'^2 \Rightarrow m_2' = m_2 + m_1'^2 \cdot \sigma, m_2' = 16 + 10^2 = 116.$$

Given $\gamma_2 = +1$

$$\sigma, \frac{m_3}{m_2^{3/2}} = 1$$

$$\sigma, m_3 = m_2^{3/2} = (16)^{3/2} = 4^3 = 64$$

$$\therefore m_3 = m_3' - 3m_2' m_1' + 2m_1'^3$$

$$\begin{aligned} \sigma, m_3' &= m_3 + 3m_2' m_1' - 2m_1'^3 \\ &= 64 + 3(116)(10) - 2 \times (10)^3 \\ &= 1544. \end{aligned}$$

$$\text{Now } \beta_2 = \frac{m_4}{m_2^2} = 4 \Rightarrow m_4 = 4 \times 16^2 = 1024$$

$$\begin{aligned} \text{and } m_4' &= 1024 + 4(1544)(10) - 6(116)(100) + 3(10000) \\ &= 92784 - 69600 = 23184 \end{aligned}$$

Comments of Nature of distribution:

Since $\gamma_2 = +1$, the distribution is moderately positively skewed, i.e., if we draw the curve for the given distribution, it will have longer tail towards the right. Further, since $\beta_2 = 4 > 3$, the distribution is leptokurtic, i.e., it will be more peaked than the normal curve.

Ex: The scores in Economics of 250 candidates appearing at an examination have

Mean $\bar{x} = 39.72$, Variance $\sigma^2 = 97.80$, Third central moment $\mu_3 = 714.18$, fourth central moment $\mu_4 = 28,396.14$. It was later found on scrutiny that the scores 61 of a candidate has been wrongly recorded as 51. Make necessary corrections in the given values of the mean and the central moment.

Ex: The following table gives the U.S. per capita consumption of whole milk (x) and of low-fat milk (y) in three different years.

	Per capita consumption (gallons)		
	1980	1984	1988
Whole milk (x)	17.1	14.7	12.8
Low-fat milk (y)	10.6	11.5	13.2

Source: U.S. Dept of Agriculture, Food Consumption, Prices, and Expenditures

Find the sample correlation coefficient r for the data.

S: To make the computation easier, let us first subtract 12.8 from each of the x values and 10.6 from each of the y values. This gives the new set of data pairs;